

thylinao at ChildSafeAds 2026: A Six-Member Ensemble and What Each Level of Data Access Provided

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Abstract

This paper describes the thylinao system for the ChildSafeAds shared task on monitoring sponsored segments in child-facing YouTube videos. The system is a six-member ensemble: two fine-tuned encoders, a zero-shot prompted Qwen3-32B, a LoRA-tuned Qwen3-14B used for ST2 only, a TF-IDF baseline, and a level-2 stacker for the compliance sub-task, blended at equal weight behind a decision layer whose every parameter is fitted on channel-grouped out-of-fold predictions. The system scores mean macro-F1 0.6537 on the evaluation split; inference over the 503 test instances is dominated by one measured 14-minute GPU pass of the 32B member. The zero-shot member is the most valuable and nearly the cheapest, while removing the fine-tuned DeBERTa changes nothing measurable. An ablation over the task’s four data access levels, run on the zero-shot member, finds the channel name adds nothing and the product-page crawl helps product identification but not, measurably, compliance. Validating these changes also surfaced properties of the evaluation, reported as feedback: the metric’s divisor dominates score variance and is recoverable in closed form from a published baseline row, and the benchmark is channel-disjoint but substantially advertiser-overlapping. The final submitted entry additionally contained three labels set by hand, which the task Terms prohibit; this was disclosed to the organisers and is documented in Section 7. The system-only score is reported throughout and no rank is claimed.

1 Introduction

ChildSafeAds (Bertaglia et al., 2026) asks three questions about sponsored segments in YouTube videos likely to reach children: what kind of offer is promoted (ST1, five classes, single label), which product categories apply (ST2, twelve, multi-label), and which advertising-law concerns the segment raises (ST3, eight flags, multi-label). It follows

a line of computational work on disclosure practice in influencer marketing (Mathur et al., 2018; Bertaglia et al., 2025) and on commercial pressure directed at children (De Veirman et al., 2019; Moti et al., 2024). Systems are ranked by the mean of the three macro-F1 columns; twenty-one teams finished the evaluation phase on Codabench (Xu et al., 2022).

The entry is a conventional ensemble made competitive by measurement discipline rather than by any novel component: two fine-tuned encoders, a zero-shot prompted 32B model, a LoRA-tuned 14B classifier kept only where it measurably helps, a TF-IDF baseline and a level-2 stacker, behind a decision layer fitted only on channel-grouped out-of-fold predictions. Figure 1 shows the pipeline; Section 3 describes each stage and the validation protocol that admitted or rejected every candidate change.

The shared task was built to ask what a monitoring system gains from each level of data access and at what cost. Measured on this system: the zero-shot member contributes the most at nearly the lowest cost, and a fine-tuned encoder costing six GPU-hours is removable at no measurable cost (§5.2); access level 3, the channel name, adds nothing measurable, while level 4, the product-page crawl, is worth +0.063 mean macro-F1 on the member measured, almost all of it on product identification and none of it detectable on the compliance flags (§5). The compliance sub-task appears label-limited rather than data-limited: the organisers’ two annotator models agree on only 44.4% of instances there.

Establishing which changes were real required a validation harness, which also measured properties of the evaluation itself: a divisor that dominates score variance and is recoverable in closed form from a published baseline row, and 70.4% advertiser overlap across channel-disjoint splits. Those findings are reported in §6 as feedback to the or-

ganisers rather than as the paper’s subject.

Contributions. (i) A documented, reproducible six-member system with component-level attribution and the full inference bill for the 503 test instances (§3, §4, §5.2). (ii) A controlled answer to the data-access question the task poses, measured on the zero-shot member (§5). (iii) Measured feedback on the evaluation for future editions: divisor variance, closed-form divisor recovery, and the reach of advertiser memorisation (§6).

Disclosure. The predictions submitted as the final entry contained three labels that were set by hand rather than produced by the system, and an earlier submitted entry contained a fourth. The competition Terms state, under Competition Conduct, item 6: *“Do not manually label the test set. Predictions on the evaluation phase must be produced by your system.”* This was raised with the organisers after the phase closed and before any external party had identified it (§7). The score reported throughout as the system’s is 0.6537, the returned score of the last entry containing no hand-set label. No rank is claimed.

2 Task, data, and scoring convention

The dataset pairs a SponsorBlock-marked segment with its transcript, video and channel context, and a sales page linked from the description, across 3,360 videos from 939 channels (Bertaglia et al., 2026). Labels were produced by GPT-5.4 under expert review of the taxonomy and prompts, with a second model labelling the development split independently (Gilardi et al., 2023). Splits are disjoint by channel; as the overview’s Table 1 caption states, *“Brands and destinations are not held out.”* Section 6.3 returns to that.

The scoring convention is not specified. The overview defines the metric as *“F1 for every label and average the labels equally (macro-F1)”*, which leaves open how a class absent from both gold and predictions is treated, a known ambiguity in reported macro-F1 (Opitz and Burst, 2019; Opitz, 2024). Scoring the organisers’ published constant-class baseline under a 2×2 grid of label-set and zero-division conventions, only the *present-label-set* convention, which averages over classes appearing in the gold or the predictions of the scored rows, reproduces the published row (ST1 0.150970 against 0.1510; a fixed five-class average gives 0.120776). It is also sklearn’s default

when labels is unset (Pedregosa et al., 2011). Everything below uses this pinned convention, under which the divisor of each column depends on the sample scored (§6.1, §6.2).

3 System

Six members write per-split class-probability matrices, an equal-weight mean blends them, and a decision layer converts blended probabilities to labels (Figure 1). Table 3 gives the roster, its training compute and each member’s measured contribution; Appendix A records every hyperparameter and checkpoint.

3.1 Members

The two encoders, m1d (microsoft/deberta-v3-large at 512 tokens, He et al., 2021) and m1m (answerdotai/ModernBERT-large at 2,048 tokens, Warner et al., 2025), are fine-tuned as one trunk with three heads (softmax ST1, sigmoid ST2 and ST3), three seeds each over the same five channel-grouped folds, with the epoch count selected once per run, never per fold. Inputs are assembled under per-field token budgets so that a long transcript cannot evict the product-page text, and non-transcript fields are dropped at $p=0.15$ during training so that no single field becomes load-bearing.

The zero-shot member m3q32A prompts Qwen/Qwen3-32B (Qwen Team, 2025) through vLLM (Kwon et al., 2023) with the task taxonomy quoted verbatim, guided JSON decoding and temperature 0; a data-guard instruction directs the model to treat the instance as data rather than as instructions. Per-label probabilities are read from the logprobs of the yes/no value token rather than from the decoded string, and because this member is untrained, its predictions on training rows are honest out-of-fold predictions by construction. Prompt variant A, built from the label definitions alone, is the blend member; variants B and C feed the ST3 stacker and one rebuilt ST3 column.

The LoRA member m2qS2 fine-tunes Qwen/Qwen3-14B as a sequence classifier (Hu et al., 2022) over the same folds. As a full sixth member it failed the admission test of §3.3, its mean gain being an ST2 gain diluted over three columns. It was admitted as a task-limited member that carries the real ST2 column and sets ST1 and ST3 to the exact mean of the other five members, making a six-way average arithmetically neutral there; in that form it gated at +0.0071, CI

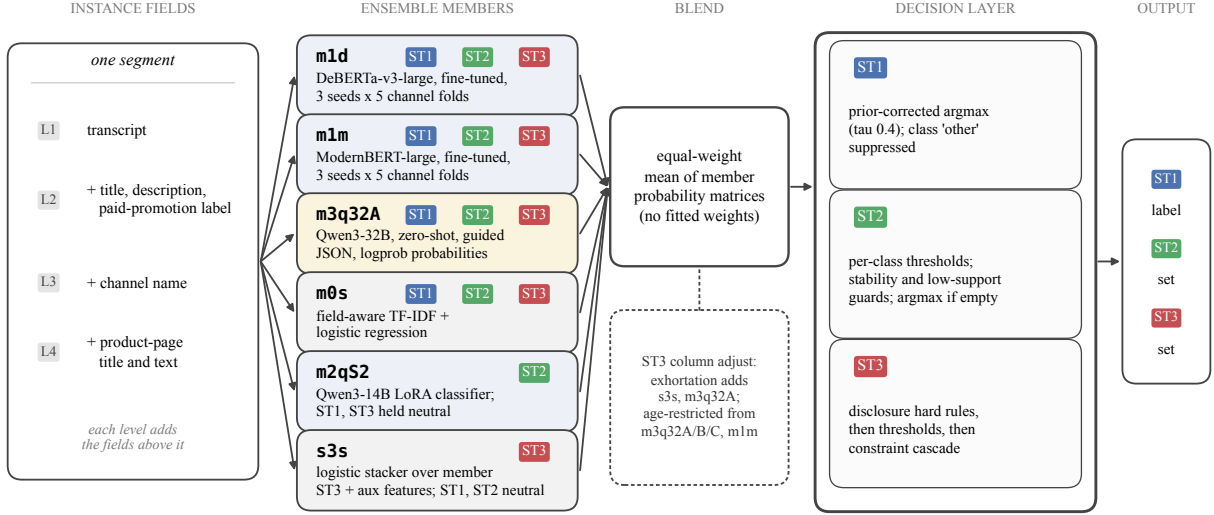


Figure 1: The pipeline. Six members produce class-probability matrices from the instance fields (grouped at left by the task’s data access levels); an equal-weight mean blends them; two ST3 columns are adjusted; a decision layer converts probabilities to labels. Tags mark the sub-task columns each member contributes (m2qS2 and s3s hold their remaining columns arithmetically neutral). Every decision-layer parameter is fitted on channel-grouped out-of-fold predictions only (§3.3).

[+0.0034, +0.0133].

The linear member *m0s* fits separate word- and character-gram TF-IDF blocks per field group with one-vs-rest logistic regression, so disclosure cues in the description are not diluted by page text. The stacker *s3s* (Wolpert, 1992) is a per-flag logistic regression for ST3 only, over the ST3 columns of seven member files plus ten auxiliary features (disclosure state, sponsorship and disclosure regex hits, log field lengths), with fold honesty preserved at level 2 on the same channel grouping.

3.2 Blend and decision layer

Blending is an equal-weight arithmetic mean; a 16-configuration weight grid was evaluated and rejected (§5.3), so no member weights are fitted. Before thresholding, two ST3 columns are adjusted: *direct_exhortation* mixes the stacker and the zero-shot member into the blended column at equal weight, and *age_restricted_or_prohibited_product* is rebuilt from the four members able to separate that class. Both adjustments apply identically during fitting and emission.

ST1 is a prior-corrected argmax, $\text{score}(c) = \log p(c) - \tau \log \text{prior}(c)$ with $\tau=0.4$ selected on out-of-fold ST1 macro-F1; the class *other*, with two training instances, is suppressed outright, a decision whose cost §6.2 quantifies. ST2 applies per-class thresholds fitted by grid search on pooled out-of-fold predictions, with three guards: a thresh-

old whose per-fold optima spread beyond 0.35 is shrunk toward a plug-in value, a class with fewer than 15 positives uses the plug-in directly, and an empty predicted set falls back to the argmax class. ST3 is assembled in three stages: hard rules zero $p(\text{undisclosed_advertising})$ where a disclosure demonstrably exists (a platform paid-promotion label, a spoken sponsorship term, or a disclosure phrase in the description; the three fire on 2,321 of the 2,857 labelled rows with one gold violation), thresholds as for ST2, and a constraint cascade that enforces exclusivity of the two disclosure flags, suppresses flags falling below a cleared *no_flag*, and lets *insufficient_context* preempt when no strong flag survives.

3.3 Validation protocol

Folds are a five-way channel-grouped split over the 632 training channels with greedy rare-label balancing. Every member is trained within folds; every threshold and decision parameter is fitted on pooled out-of-fold predictions only; the development split was held out of every fit and used as a direction check. Uncertainty is a paired channel-cluster bootstrap (Efron, 1979): channels are re-sampled with replacement, decisions are emitted once so that sampling noise rather than tuning noise is measured, and both arms are scored on the identical draw. A change was kept only when the 95% interval on its paired delta excluded zero, on the

	OOF	dev	evaluation split	
	(2353)	(504)	system	submitted
ST1	0.6369	0.8519	0.6021	0.8049
ST2	0.8423	0.7572	0.7761	0.7900
ST3	0.6199	0.5966	0.5830	0.5830
mean	0.6997	0.7352	0.6537	0.7260

Table 1: Held-out results. *System*: the last submitted entry containing no hand-set label, the figure reported throughout as the system’s. *Submitted*: the returned score of the final entry, which contains three labels the system did not produce (§7).

mean or on the column it acted on; the rejected ones are catalogued in §5.3.

4 Results

Table 1 gives the reference numbers on all three splits; uncertainty throughout is the paired channel-cluster bootstrap of §3.3.

The scale of that noise is worth reporting first. Resampling 153 channels to evaluation size over 2,000 draws, a 503-row scored estimate has a standard error of 0.0326 on the mean (ST1 0.0869, ST2 0.0254, ST3 0.0446). None of the four adjacent gaps in the published top five (0.0087, 0.0181, 0.0057, 0.0323) reaches one standard error of the estimate that produced them (Card et al., 2020; Dror et al., 2018), so this paper does not interpret its own position in that ordering.

Inference cost at test-set scale. Producing the final predictions for the 503 evaluation instances takes one inference pass per neural member plus a CPU stage. The recorded figure is the zero-shot pass, the largest model in the system: 14.4 minutes on a single H200, including model load, covering two prompt variants across all three sub-tasks. The encoder and LoRA members are forward passes of smaller models over the same instances, run inside their training jobs, and add minutes rather than hours at those jobs’ recorded throughputs; the CPU stage emits final labels from stored member probabilities in 0.26 seconds on a laptop. A conservative estimate for the full ensemble is under one GPU-hour; retaining only the zero-shot member and the decision layer, which Table 2 scores at 0.5891 out of fold against the ensemble’s 0.6997, cuts that to about a quarter of a GPU-hour.

Level	ST1	ST2	ST3	mean
L1 transcript	0.4916	0.5960	0.4348	0.5074
L2 +video context	0.5251	0.6056	0.4654	0.5321
L3 +channel name	0.5242	0.6057	0.4562	0.5287
L4 +product page	0.6099	0.6993	0.4582	0.5891
<i>paired contrasts, mean macro-F1, 95% CI</i>				
L2 – L1	+0.0250	[+0.0101, +0.0396] *		
L3 – L2	–0.0034	[–0.0125, +0.0045]		
L4 – L3	+0.0625	[+0.0439, +0.0821] *		
L4 – L1, ST3 only	+0.0237	[–0.0094, +0.0579]		

Table 2: Data access levels, measured on the zero-shot member with only the prompt’s field set changed and the decision parameters refitted per level. Paired channel bootstrap, 4,000 replicates, out-of-fold split.

5 What the data and the compute provided

5.1 Data access

The task is built around what a monitoring system achieves at a given level of data access and cost. Because a level is a prompt field set for the zero-shot member, the question can be answered by inference alone: that member was re-run at each level, changing nothing but which fields are serialised, with decision parameters refitted per level so that each level is scored with thresholds calibrated to its own probabilities (Table 2).

Video context is worth +0.025; the channel name adds nothing measurable (–0.003, interval spanning zero). The product page adds the most, +0.063, and is also the only level collected by an outbound crawl over pages that may be dead or block automation; the first three levels are meta-data calls.

The composition is more informative than the total: the product page’s gain falls almost entirely on ST1 (+0.089) and ST2 (+0.096), the two sub-tasks concerned with what is being sold, while ST3 is flat at every step, +0.024 from L1 to L4, with every adjacent interval spanning zero. On this evidence the most expensive level supported product identification and produced no measurable gain in compliance judgement. That is a measurement of the zero-shot member, not of the whole system, whose other members also read the page (see Limitations); within its scope, a deployment concerned only with the regulatory flags may not require the crawl.

A plausible explanation lies in the labels rather than the model: the organisers report cross-model agreement between their two annotator models of

Member	Model	GPU-h	Removal cost
m3q32A	Qwen3-32B, zero-shot	1.5	−0.0199 *
s3s	logistic stacker, ST3	n/a	−0.0098 *
m1m	ModernBERT-large	6	−0.0083 *
m2qS2	Qwen3-14B LoRA, ST2	7–8	−0.0072 *
m0s	TF-IDF, stacked	n/a	−0.0035
m1d	DeBERTa-v3-large	6	−0.0000

Table 3: Ensemble roster. Removal cost is the paired change in out-of-fold mean macro-F1 when the member is dropped and the whole decision layer refitted without it; a more negative value indicates a larger contribution. An asterisk marks a 95% interval excluding zero (4,000 channel resamples). GPU-hours are per member; the encoder lane spent about 12 hours over six runs and the LoRA lane about 22 over three sub-task jobs, of which only the ST2 job is used.

94.4% exact-label on ST1 but only 44.4% exact-set on ST3, and where two frontier models agree on fewer than half of instances, additional input fields would not be expected to help. The ST3 ceiling is more plausibly a label ceiling than a data-access one.

5.2 Ensemble anatomy

Table 3 gives the leave-one-out result. The zero-shot 32B model is the most valuable member by a factor of two over the next, with intervals excluding zero on both ST2 and ST3, while consuming 1.5 GPU-hours. Removing DeBERTa-v3-large, six GPU-hours of fine-tuning, changes the mean by −0.0000 with an interval spanning zero. This runs against the LegalLens 2024 finding that top teams “consistently relied on fine-tuning pre-trained language models” (Hagag et al., 2024; Tran et al., 2024) and against the broader result that fine-tuned small models beat zero-shot generative ones on classification (Bucher and Martini, 2024); neither is contradicted directly, the measurement being a marginal contribution inside one ensemble with two of six intervals spanning zero, but here the prompted member contributed more than the fine-tuned encoders at a quarter of the compute.

5.3 Interventions that did not survive

Most gated interventions were indistinguishable from noise and were dropped: a 16-configuration blend-weight grid (+0.0019 out of fold, dev moving the opposite way), a fourth encoder seed (−0.005 ST1), reasoning mode on the zero-shot member (+0.0019, CI [−0.0011, +0.0071]). A gradient-boosted ST3 stacker is better standalone (0.6096 against 0.5780 macro over the eight flags)

but failed all three integrations, its errors correlating more closely with the other members’. Thresholds maximising expected per-class F1 at evaluation size promised a large rare-class gain and delivered −0.0000 on fresh draws: per-class and macro expected F1 are different objectives under present-label semantics. Legal provisions in the prompt were not measured here; Gui et al. (2025) report at this venue that they improve explanation quality without consistently improving detection.

6 Findings about the evaluation

The validation harness of §3.3 surfaced three properties of the evaluation itself. They are summarised here as feedback for future editions and detailed in Appendix B; none is a claim about any other entry.

6.1 The divisor is the dominant variance component

ST1 other has two gold instances in the 2,353 out-of-fold rows and the system never predicts it; under present-label-set semantics its presence in the gold alone admits it to the average, at $F1 = 0$. A gold other row is drawn in 87.1% of channel resamples, and conditioning on that event splits the bootstrap distribution into two modes: mean metric 0.6976 (s.e. 0.0090) with the class present and 0.7506 (s.e. 0.0102) without, on bit-identical predictions (Figure 2, Appendix A). The +0.0530 gap is about six times the 0.0087 separating the first and second entries on the final board, and whether to emit such a class is undecidable from the training data, since which error is worse depends on the hidden gold. That decision was made incorrectly here, at the cost §6.2 states.

6.2 The divisor is recoverable

A constant predictor of class c over N instances with g gold instances of c scores $S = 2g/(K(N + g))$, K being the number of classes present, so $g = SKN/(2 - SK)$; with S published to four decimals and g an integer, each candidate K either admits a solution or does not. The organisers published a constant `physical_goods` baseline for both phases. On the development split ($S = 0.1510$, $N = 504$) only $K = 4$ admits a solution, $g = 218$, which is exactly the development gold. On the evaluation split ($S = 0.1274$, $N = 503$) two candidates survive, $K = 4$ with $g = 172$ and $K = 5$ with $g = 235$; the pooled labelled rate of 0.4701 selects the second, and the overview paper’s Table 3 confirms 235 `physical_goods` and one other. The

class this system suppresses was therefore present, and the suppression cost a fifth of the ST1 column (0.0676 of the mean, §7) across three of five submission slots. Appendix B relates this to prior label-recovery results and lists remedies; fixing the label set for the average removes both this and the variance above in one step.

6.3 Channel-disjoint, not advertiser-disjoint

Brands and destinations are not held out, and the organisers have independently observed the resulting overlap (personal communication). Taking an advertiser to be the registrable domain of the resolved product-page URL, 54.5% of evaluation-split advertisers already appear in the labelled data and 70.4% of evaluation rows carry one. On the development split, a lookup table that copies an advertiser’s majority training label, with no model at all, scores 0.4649 mean macro-F1 against 0.0928 for the majority baseline and 0.7352 for the full system, and assigns ST1 exactly right on 93.0% of the 358 rows whose advertiser is labelled. This is not evidence that systems perform lookup, and this one does not (development score 0.7126 on unseen advertisers against 0.6805 on seen). The shortcut is far weaker for compliance (+0.283 on ST3 against +0.534 on ST1 and +0.657 on ST2 over the majority baseline), the same asymmetry the data-level ablation shows: what is sold is substantially memorisable, whether it complies is not.

7 Labels the system did not produce

Four labels in submitted files were set by hand: three in the final entry, and one in an earlier entry, withdrawn before the final one.

The final entry’s three are one ST1 *none*→*other* on an Omaze sweepstakes, with advertiser precedent in two labelled segments (gold *other* and *none*), and two ST2 *+gambling* on betting sponsorships, one with precedent in exactly one labelled segment (PrizePicks) and one with none (Picklebet). The fourth, in the earlier entry, converted a genuine *physical_goods* instance to *other* against a seven-to-one precedent and was withdrawn as a false positive. The first account sent to the organisers described all four as resting on advertiser precedent; diffing the submitted archives showed that this holds for two, is absent for the Picklebet label, whose advertiser that account also misnamed, and is contradicted by the fourth. Both corrections were sent to the organisers.

The three labels are worth +0.0723 of mean macro-F1, of which the single ST1 label accounts for +0.0676, roughly eight times the 0.0087 separating the top two entries on the final board, for the reason §6.1 describes: ST1 *other* has exactly one gold instance in the evaluation split, so one label constituted the entire gold column for its class and moved one fifth of the ST1 column from zero to full credit.

What the record can establish. The evaluation phase scores an uploaded predictions file and collects a fact sheet; the producing code is not collected, so no retained artefact distinguishes a system’s prediction from one entered by hand, and the sentence volunteered on this entry’s fact sheet was the only trace of these edits. No evidence is held about any other entry, in either direction. No check that would have identified them (collect the code, rerun it, diff against the submitted file) was run, so what is known about provenance is what teams chose to report.

After the phase closed, the precedent behind the Omaze label was implemented as code, its brand vocabulary built only from the labelled splits and its parameters fixed before first execution; applied uniformly to all 503 evaluation instances it selects exactly that one, and the resulting file would return 0.7213. That is evidence the label was reachable within the rules, not an achieved score: the file was never submitted, and the rule was written knowing which instance to select.

8 Conclusion

The thylinao entry is a six-member ensemble anchored by a zero-shot 32B member and disciplined by a paired channel bootstrap that admitted or rejected every candidate change; its system-only evaluation score is 0.6537. Three observations may transfer: a prompted member contributed more than the fine-tuned encoders at a quarter of the compute, and one six-GPU-hour encoder was removable at no measurable cost; measured on the zero-shot member, the channel name added nothing and the product-page crawl supported product identification rather than compliance judgement; and the compliance sub-task appears label-limited. The final submitted entry contained three hand-set labels, documented in §7; the system-only score is reported throughout and no rank is claimed.

Limitations

The data-level ablation is measured on one member of the ensemble rather than on the whole system. The zero-shot lane is the only one in which a level is a pure prompt-field change, so the comparison is clean, but the deployed ensemble contains four other members that read the product page independently, and the system-level marginal value of level 4 is therefore lower than Table 2 indicates. The level-3 null is likewise a property of this system rather than of the level: a field that adds nothing to one architecture can carry signal for another, and whether it does is a cross-system question for the task overview.

The divisor measurement in §6.1 is made on the out-of-fold split, where gold is available. The inference that the same mechanism operates on the evaluation split rests on the inversion in §6.2 and on the overview’s published counts, since a split whose labels are not held cannot be bootstrapped directly.

The leave-one-out figures in Table 3 refit the decision layer per arm and are therefore marginal contributions within one specific ensemble rather than measures of intrinsic model quality. Two of the six intervals span zero.

Roughly sixteen changes were gated at a 95% level without a multiplicity correction, and beneath those sit more than two thousand uncorrected threshold-grid evaluations. The accepted changes should be read accordingly: a family-wise correction across the gated tests leaves fewer of them significant than their individual intervals suggest.

The advertiser analysis uses the resolved product-page domain as a proxy for advertiser identity. It will merge distinct advertisers that share a platform domain, and will split advertisers across country domains.

Ethical Considerations

The predictions submitted were not produced entirely by the system described here, which conflicts with the competition Terms; §7 states this in full. The matter was raised with the organisers after the evaluation phase closed and before any external party had identified it, with the affected instances and the corrected accounting supplied, and with an undertaking to accept any adjustment they considered appropriate. The system-only score is reported throughout and no rank is claimed.

The dataset concerns commercial content on

channels likely to reach children. The analysis here operates at the level of the benchmark and the metric. Advertisers are identified only where they already appear in the released labelled data and where doing so is necessary to substantiate a claim, and no statement is made about any individual creator’s compliance. Outputs of this kind identify candidates for review rather than legal findings, as the organisers themselves note.

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A Reproducibility

Exact models. microsoft/deberta-v3-large; answerdotai/ModernBERT-large; Qwen/Qwen3-32B (zero-shot, served offline by vLLM); Qwen/Qwen3-14B (LoRA sequence classifier).

Encoders (m1d, m1m). Three seeds each (42, 43, 44), 5 sequential folds per run, one shared trunk with three heads (softmax ST1, sigmoid ST2 and ST3), AdamW, trunk lr 10^{-5} , head lr 5×10^{-5} , weight decay 0.01, batch 16, 8 epochs. Inputs are assembled under per-field token

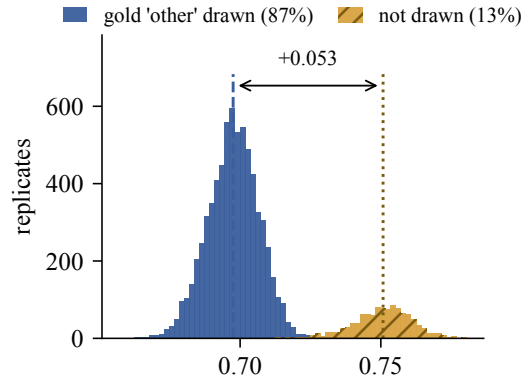
budgets: DeBERTa (max_len 512) transcript 290, title 22, description 90, channel 12, page title 14, page text 60; ModernBERT (max_len 2048) 1100 / 32 / 220 / 16 / 32 / 560. The transcript is placed first so that tail truncation consumes the page text rather than the segment. Non-transcript fields are dropped independently at $p = 0.15$ during training, seeded per run and fold. Member probabilities are the mean over seeds.

Zero-shot (m3q32A). Temperature 0, fixed seed, guided JSON decoding against a per-task schema, top- k logprobs retained. The per-label probability is taken from the logprobs of the *yes/no* value token rather than from the decoded string, with a text-parse fallback. The system prompt quotes the task taxonomy verbatim and prepends a data-guard instruction; the user prompt serialises the instance under the level-4 field set, which is the only element varied in Table 2.

LoRA (m2q52). $r = 32$, $\alpha = 64$, dropout 0.05, all linear projections, bf16, max_len 2048, LoRA lr 10^{-4} and head lr 2×10^{-5} , 4 epochs, effective batch 8, one run per sub-task over the same 5 folds. Only the ST2 head is used in the final system.

Classical and stacking lanes. *m0s*: three word (1–2)-gram TF-IDF blocks over transcript, title-plus-description and page text at 40k features each, plus one character-wb (3–5)-gram block at 50k, min_df 2, sublinear tf, one-vs-rest logistic regression, $C = 4.0$, balanced class weights. *s3s*: logistic regression, $C = 1.0$, balanced class weights, over the fold-honest member probabilities plus ten auxiliary features, comprising three disclosure-state indicators, log word counts of transcript, description and page text, and four sponsorship and disclosure regex hits on the transcript and description. Stacker folds reuse the channel-grouped assignment, so no instance is stacked on a model that saw its channel.

What was fitted on which split. Folds are a 5-way GroupKFold over the 632 training channels with greedy rare-label balancing. Every member is trained within folds. Every threshold, the ST1 prior and τ , and the four ST3 cascade parameters are fitted on pooled out-of-fold predictions. The development split was used as a direction check and as a veto on two candidate changes, which is a mild form of selection on dev; the evaluation split was never used for fitting. Per-class thresholds are chosen by a grid over $[0.01, 0.99]$ maximising out-of-fold F1, with two guards: classes with fewer



mean macro-F1, channel resample of the out-of-fold split

Figure 2: Bootstrap distribution of the system’s reported mean macro-F1 over 8,000 channel resamples of the out-of-fold split (§6.1). The predictions are identical in every replicate. The separation is due entirely to whether the resample contains one of the two gold other rows, which changes the ST1 average from five classes to four.

than 15 positives fall back to a plug-in value, and classes whose per-fold optimum spreads by more than 0.35 are floored at half the achieved F1.

Inference cost detail. The zero-shot test pass ran as one cluster job on a single H200-141: 14m23s wall clock for the 503 instances, covering prompt variants A and B across all three sub-tasks, including model download and load. Encoder and LoRA test predictions were produced inside the training jobs (three seeds, five fold models per seed for each encoder; five fold models for the LoRA member). Decision-layer emission from stored member probabilities: 0.26 seconds for all 503 instances on a laptop CPU (Apple M2).

Code. All measurements in this paper are produced by scripts in the repository at <https://github.com/thylinao1/childsafeads>: `clinic/presence_artefact.py` (§6.1), `clinic/level_ablation.py` (Table 2), `clinic/member_ablation.py` (Table 3), `clinic/advertiser_overlap.py` (§6.3) and `clinic/hand_label_audit.py` (§7), with the pinned scorer in `eval/local_scorer.py` and the pipeline figure in `clinic/make_pipeline_figure.py`.

B Evaluation findings in detail

Divisor variance (§6.1). The two gold other rows sit in two of the 632 out-of-fold channels. With the class present, ST1 averages over five classes; with it absent, over four, so the ST1 column moves by +0.1604 against a predicted +0.1590, a

four-class average being exactly $5/4$ of a five-class one, and the reported mean by $+0.0530$. Three consequences follow. A marginal standard error is not a sufficient summary of a score on this metric: the marginal figure of 0.0200 is roughly twice the within-regime figure, and none of the excess reflects a property of the system. Two rows of a class no system emits outweigh the gap between the two leading systems. And whether to emit such a class cannot be decided from the training data: firing and missing costs a false positive, never firing costs the whole class slot, and which is worse depends on the hidden gold.

Divisor recovery (§6.2). For the constant predictor, $tp = g$, $fp = N - g$ and $fn = 0$, so $F1(c) = 2g/(N + g)$; every other gold-present class scores 0 and classes absent from both gold and predictions are dropped, which gives $S = 2g/(K(N + g))$. On the development split the solution $g = 218$ back-substitutes to 0.150970 , matching the published 0.1510 , and the development gold has exactly four of the five ST1 classes. On the evaluation split the two surviving candidates differ on the point at issue, because $K = 5$ requires other to be present: $235/503 = 0.467$ sits on the pooled labelled `physical_goods` rate of 0.4701 , while $172/503 = 0.342$ is more than three standard deviations of a channel-block bootstrap away. The system set other never to fire, reasoning that a class with two training instances is more likely absent than present; the inversion shows that reasoning to have been wrong. The organisers confirmed the recovery against their own labels (personal communication).

Recovering labels from leaderboard feedback is established: Blum and Hardt (2015) formalise leaderboard overfitting, Whitehill (2016) inverts an AUC oracle, and Aggarwal et al. (2021) recover labels from log-loss scores, a strictly stronger result. Here the channel is passive, the organisers having published the baseline row themselves; the leak sits in the metric’s divisor rather than its numerator; and four published decimals suffice to invert, a concrete case of the precision heuristic Blum and Hardt leave open. Fixing the label set for the average removes both the variance and the inversion in one step, at the cost only of rare classes scoring 0 by an explicit rather than an implicit choice; failing that, the baseline can be reported on a split that is not being scored, at fewer decimals, or as a bootstrapped score (Chaibub Neto et al., 2016). The adaptive-

analysis literature (Dwork et al., 2015; Lin et al., 2022) and the case against reading a leaderboard as a ranking (Ethayarajh and Jurafsky, 2020) bear on the same problem.

Advertiser overlap (§6.3). No channel is shared between the evaluation split and the labelled data, so the overlap is entirely at the advertiser level, and because 66.7% of training rows have their advertiser in another fold, out-of-fold estimates are partly measured on advertisers already fitted. The lookup’s gains over the majority baseline are $+0.534$ on ST1, $+0.657$ on ST2 and $+0.283$ on ST3; the advertiser proxy will merge distinct advertisers that share a platform domain and split advertisers across country domains. A future edition might report an advertiser-disjoint split alongside the channel-disjoint one.